

WRRRA Prime-Parts Rendering: Finite Assembly, Neutrality, Composition and Decay

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Abstract and evaluation ledger

This exploratory synthesis studies a finite arithmetic rendering rule for odd composite addresses admitted by a phenotype filter. Upstream multiplicative prime factorization and downstream additive rendering are explicitly different operations. Thirteen computational probes are reviewed together with an earlier fold-current beta-decay calculation. The results establish order-dependent arithmetic composition, conserved signed-charge ledgers, conditional agreement with a coarse hydrogen-helium composition target, and an executable capture-decay competition. They do not yet establish an address-to-rest-mass constitutive law or a production Hamiltonian.

Verified inputs: recorded particle masses, neutron mean lifetime, charge assignments, and a coarse 75:25 hydrogen-helium mass benchmark. The finite filter exponent and admission coefficient are inherited calibration inputs, not measured particle data.

WRRRA transformation: finite weighted address pool → additive prime parts and residual → declared particle motifs → quark and charge ledger → conditional decay and capture.

Outputs: assembly-order total variation approximately 0.204768; a fixed decoder with the first 16 primes gives a conditional neutron fraction approximately 0.124839; a high-capture case yields a helium baryon-count proxy approximately 0.247207.

Falsification conditions: failed arithmetic or charge conservation, unreproducible outputs, an energetically impossible physical decoder, or composition mismatch after the chosen constitutive rules and calibration are held fixed. Reproducing verified values is treated as explanatory consistency; independent prediction is not a prerequisite for that achievement.

1. Scope, hypotheses and provenance

The research goal is a nonprivileged single universe described through minimal computation, a common carrier and phenotype rendering. This paper is a separate parts-research branch; it does not revise the established upstream/downstream synthesis. All calculations use finite address cutoffs. No physical infinite limit is required.

The working hypothesis separates unsigned load from signed or directional deformation. Only electric charge is implemented as a signed scalar in the present ledgers. Other polarities require specified operators and cannot be inferred from electric neutrality.

The phrase pseudo-prime is operationally restricted here to admitted odd composite addresses, not a claim that every address is a pseudoprime in a named primality test. A confused-prime decoder has not been supplied by the filter. The candidate reference-prime rules are declared diagnostic decoders.

The heaven/earth-stem analogy motivates the distinction between abstract factorization and phenotype assembly. It supplies neither a numerical constraint nor proof. Neither 15 prime labels nor 16 prime labels are identified with carrier-channel counts merely because their integers coincide.

Author: Wonsik Choi (ORCID 0009-0001-4263-9772). This release has one author. Earlier reference studies retain their original provenance; they are not newly authored versions of those studies.

2. Finite filter and reference-residual split

Let A_N contain odd composite integers n from 2 through N . Set $Z_N = \sum (n^{-\alpha})$, with the sum over all integers 2 through N . The admitted weight is $a_n = \beta * n^{-\alpha} / Z_N$. At $N=1,000,000$, $\alpha=1.8996876950554356$ and $\beta=0.8654570124136961$, A_N contains 421,502 addresses and the admitted weight sums to approximately 0.05. Conditional weights are $w_n = a_n / \sum(a_n)$.

Two reference rules choose the previous odd prime or the nearest odd prime, with ties resolved downward. The signed split $n=p+\Delta$ is exact. The diagnostic pair count $k=\text{abs}(\Delta)/2$ is an extra rule: it is an integer for odd n and odd p , but does not itself create particles or assign rest energy.

The previous-prime rule gives conditional mean k approximately 1.311067 and maximum 56; the nearest-prime rule gives mean approximately 1.071290 and maximum 28. Reference/count outcomes change on 194,071 addresses. Thus the numerical meaning of the split depends on the decoder.

The prime p is a reference label, not automatically an emitted prime particle. Negative Δ is a signed residual and is not negative physical energy.

3. Additive assembly and order effects

For a pool P_K of the first K primes, write $n = \sum(c_j * p_j) + r$, with nonnegative integer counts and $r \geq 0$. The one-each cyclic algorithm scans each prime once per cycle, subtracts it when it fits, and repeats until no prime fits. Ascending and descending scans use the same pool and address weights. Complete cycles may be accelerated by quotient subtraction because every member of a complete cycle fits.

Maximal-repeat rules are separate alternatives. Ascending maximal repetition uses only 2 and leaves residual 1 on odd inputs. Reserving 3 before repeating 2 eliminates that residual. This demonstrates that a completion criterion materially changes the parts profile.

For $K=15$, ascending one-each assembly has conditional mean 15.969611 parts and descending assembly 11.427223. Conditional residual-1 weights are approximately 0.169178 and 0.162073. Exact-address counts are 253,697 and 280,316, respectively. The component-number distributions have total variation approximately 0.204768.

Conservation $n = \sum(c_j * p_j) + r$ holds for every tested address. It proves the arithmetic implementation, not mass conservation. Neither order is universally selected: changing K or the completion criterion changes which outcome appears preferable.

4. Charge motifs, calibrated composition and cutoff stability

Each component can be assigned either a neutral proton–electron motif or a neutron motif. Under this declared decoder, $N_p=N_e$ by construction. Opposite–charge pair generation also preserves net charge while leaving species ratios unresolved. These are executable realization rules, rather than a derivation that global neutrality alone forces proton–electron equality in every universe.

The coarse 75:25 H/He mass target, neglecting binding energies and mass differences, corresponds to 12 hydrogen atoms per helium–4 atom and $p:n:e=7:1:7$. It is a baryonic composition benchmark, not a count ratio for all photons, neutrinos or dark components.

All 32,766 nonempty proper binary subsets of a 15–label pool were inspected. A joint calibration selects neutron labels {5,47}; it gives $f_n=0.129335$ for ascending and 0.127097 for descending assembly. This is a valid calibrated explanatory result. Its subset–search provenance is retained rather than represented as a unique mandatory decoder.

Keeping {5,47} fixed, 56 cutoff/order cases were evaluated: N in {1,000,10,000,100,000,1,000,000}, K in {2,4,8,12,15,16,20}, and two orders. Missing labels are not replaced. At $N=1,000,000$, ascending f_n for $K=8,12,15,16,20$ is approximately 0.126533, 0.098775, 0.129335, 0.124839, 0.113958; descending values are 0.116939, 0.071447, 0.127097, 0.111249, 0.081866. $K=16$ ascending is a useful candidate, not a uniquely derived minimum. Its f_n changes only about 0.0000441 from $N=100,000$ to 1,000,000 in the tested finite range.

5. Load conservation and the failed pair-energy shortcut

A declared linear load map $E(n)=\epsilon n$ and $E(r)=\epsilon r$ preserves the additive ledger. Calibrating the whole admitted sector to 5 percent yields, for ascending $K=15$, parts 4.996442 percent plus residual 0.003558 percent; descending gives 4.996591 plus 0.003409 percent. The conditional mean address load is approximately 237.731551 in both cases. These fractions use the full-universe denominator and are not particle-number shares.

This linear load map is a new constitutive assumption. It is not derived from the earlier phase-state Hamiltonian. In the earlier equal-occupation spectral state, a diagonal Hamiltonian gives the same energy expectation approximately 402,544.971 eV at every address. Therefore the difference between address expectations is zero and cannot pay for positive-mass pair production.

A state-distance diagnostic on 4,096 deliberately selected addresses changes 2,250 reference choices relative to the previous-prime rule. The sample is not unbiased. Common unitary-frame transformations preserve its distances to numerical precision. A proposed bounded distance-based budget has maximum approximately 15,213.258 eV, below the electron-positron rest threshold approximately 1,021,997.901 eV for a one-packet interpretation.

This failure rejects that specific energy-funding shortcut. It does not reject all additive rendering maps. A physical rest-mass allocation must specify units, energy reservoirs, binding contributions and open outputs without double counting pair energies.

6. Quark assembly, neutrality and open channels

Valence bookkeeping uses $p=uud$ and $n=udd$. The normalized antisymmetric RGB tensor is a color singlet: the eight total color generators annihilate it. This is a color check; the current parts branch does not solve a QCD binding Hamiltonian or a complete spin-flavor wavefunction.

A controlled inventory has $u=15$, $d=9$ and $e=7$, with each color carrying $u=5, d=3$. Under a restricted uud/udd greedy assembly, proton-first gives $p=7, n=1, e=7$ with no residue. Neutron-first gives $p=3, n=3, e=7$ and six unresolved u valence units. Charge is conserved only when the residual ledger is included. The residual is not claimed to be six observable free quarks.

Allowing color-singlet-compatible uuu, uud, udd, ddd inventories gives 193 charge-neutral allocations, including 11 using all valence units. Global neutrality therefore does not select a unique p/n composition. Spin, energy and accessible reactions provide further restrictions.

Uniformly holding back hypothetical fractions 0, 0.05 or 0.20 into open branches leaves the internal $p/n/e$ ratios unchanged; nonuniform branch allocation need not do so. Neutrinos, photons and unresolved residuals remain explicit outputs. The hypothesis that unused parts become CMB photons remains open: a photon count or residual integer does not produce a 2.72548 K blackbody or its measured energy density.

7. Decay: earlier fold-current calculation and current ledger

The earlier fold-decay 0.2 already implements a stronger connection than decay counts alone. Declared u/d prime markers 3 and 5 give proton address $45=3^2 \cdot 5$ and neutron address $75=3 \cdot 5^2$. In an explicit symmetric three-quark spin-flavor sector with antisymmetric color, the computed vector matrix element is 1 and the bare axial element is $5/3$.

The finite local Euler-family response is $r_p = \sum (p^{(-\alpha \cdot k)})$ over $p^k \leq N$. The transition amplitude uses a declared common kernel $\kappa \sqrt{r_3 \cdot r_5}$. Observed axial ratio fixes $\eta_A = 0.76518$; one neutron lifetime fixes κ approximately 12.200295. Particle masses give Q approximately 0.782333559 MeV. A Coulomb-corrected allowed beta phase-space integral gives approximately 1.69166524.

The calibrated rate returns 878.3 s, while fixed-kernel $Q=0.25, 0.5, 1, 2$ MeV cases yield approximately 65,897.154, 4,974.762, 330.233, 18.502 s. The mean electron kinetic energy is approximately 0.301448 MeV in the stated leading approximation. The Lindblad/Kraus transport preserves trace, positivity, charge, baryon number and total Actual energy. All 30 prior checks pass on review. Thus a fold-current/rate execution exists; it should not be described as wholly unimplemented.

These markers are multiplicative upstream labels. The additive decoder {5,47} is not the same map: no equality between a neutron component label 47 and the nucleon address 75 is asserted. The exact interface between the two address semantics remains to be specified.

The current 11-allocation decay ledger conditionally identifies uuu and ddd with Δ^{++} and Δ^- resonances, subject to the necessary spin/excitation state. Supplied pion/muon chains and assumed two-photon annihilation conserve charge and baryon number, with nonnegative available nonrest budgets. Individual energy spectra and rates for these chains are not computed. If every remaining neutron is free and allowed to decay, the limiting count inventory is $p=8, n=0, e=8$, rather than 7:1:7.

8. Capture-decay competition and corrected interpretation

Set the initial conditional neutron fraction $f=0.12483936387544717$. Let $d=1/878.3$ per second and k be a prescribed effective capture rate. For exposure T , $\text{free}=f*\exp(-(k+d)*T)$, $\text{captured}=f*k/(k+d)*(1-\exp(-(k+d)*T))$, and $\text{decayed}=f*d/(k+d)*(1-\exp(-(k+d)*T))$. Their sum is f . Proton and electron counts are $1-f+\text{decayed}$, and baryon counts remain one.

The proxy $Y=2*\text{captured}$ assumes that every captured neutron ends in helium-4 with an equal number of protons. It is a baryon-count mass approximation, not an exact nuclear mass fraction or a reaction-network prediction. The assumed capture sink does not establish helium formation.

At $T=180$ seconds, $k/d=1, 10, 100$ gives Y approximately 0.041980, 0.203161, 0.247207. The no-loss ceiling is $2*f$ approximately 0.249679, so exact 0.25 is unavailable with the fixed initial f . Rough agreement is consistent with the intended coarse benchmark. The 180-second window and capture ratios are assumptions, not a WRRRA-derived cosmic clock or measured reaction rates.

The result demonstrates a composition degeneracy: initial arithmetic mixture and later capture history can compensate each other. It is therefore inappropriate to choose an assembly order using a final H/He ratio without an evolution rule. A temperature/density-dependent nuclear network is the next physical extension.

9. Review, conclusions and next constraints

All 13 original studies were executed again. Nine result JSON files were exactly equal; four differed only in floating-point rounding and all 13 agreed at relative tolerance $1e-11$ or absolute tolerance $1e-12$. Earlier fold-decay 0.2 executed successfully and passed 30/30 checks. Source archives, reviewed code, input snapshots, replay logs and numerical outputs accompany this release.

Corrections in this synthesis: distinguish conditional number weights from energy fractions; call the helium output a baryon-count proxy; retain decoder calibration provenance; distinguish global neutrality from pair equality; keep unresolved colored inventories out of observable-particle claims; recognize the existing fold-current rate calculation; distinguish additive parts labels from multiplicative nucleon labels; and preserve the failed equal-expectation pair-energy shortcut.

The explanatory achievement is an executable finite chain from weighted admitted arithmetic addresses to order-dependent component profiles and consistent charge/evolution ledgers, with a calibrated composition candidate. The open physical interfaces are specific: a component-to-species law with spin and color, a shared dimensional rest-energy/binding ledger, and capture rates under a defined expansion/thermal history.

Future work should first test the mass-energy interface, then connect the earlier fold-current transition to the additive renderer, then implement physical capture reactions. Once calibrated constitutive rules are fixed, their unmeasured outputs may be treated as WRRRA predictions. Known-value reproduction already supplies a legitimate reality-consistency result within the declared scope.

References and reproducibility

[1] W. Choi, WRRRA M Fold-Current Beta-Rate Bridge 0.2, source revision 43b998ca0ac63c6ae21eb0e0bc543e98172a42e4; upstream/fold_decay_v0_2 in <https://github.com/Wonsik-Choi-janefather/wrra-m-0.1> .

[2] J. S. Nico, Neutron beta decay, J. Phys. G 36 (2009) 104001.
https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=902353 .

[3] NASA, Imagine the Universe, cosmology reference.
https://imagine.gsfc.nasa.gov/ask_astro/cosmology.html . The 75:25 target is used only as a coarse H/He benchmark.

[4] NASA LAMBDA, CMB and Radio Monopole Spectrum.
https://lambda.gsfc.nasa.gov/resources/lambda_graphics/cmb_monopole.html .

[5] Particle Data Group, particle mass and quark-model reference inputs, as recorded in the archived study inputs and earlier fold-decay code. No new precision fit is claimed.

Run python reproduce.py from the release root. Requires Python 3, numpy and scipy. Replays are compared numerically to stored baseline outputs. The earlier reference study is included as a provenance snapshot, not silently merged into the parts renderer.

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